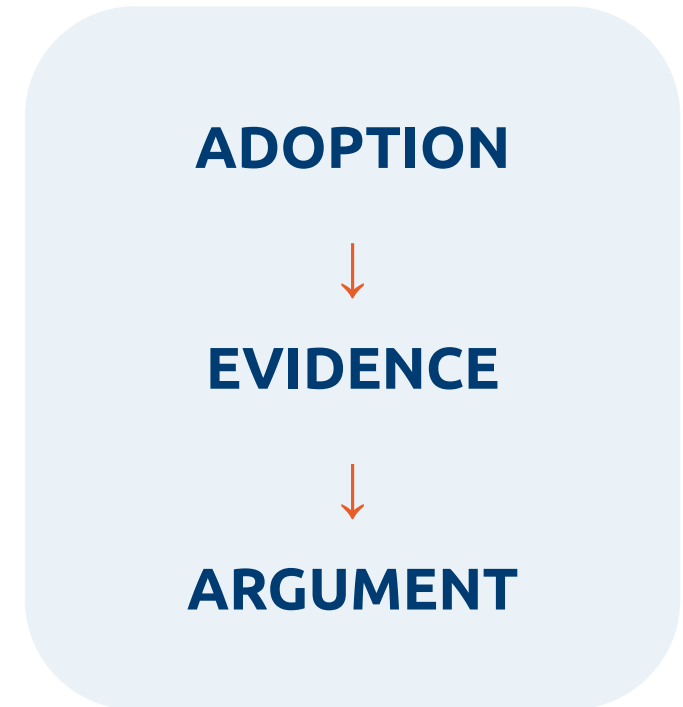


MODULE 5 TUTORIAL

Adoption & Evidence-Based Arguments

How technologies spread and how recommendations earn support



Two questions for today

TECHNOLOGY & SOCIETY

How do new technologies become part of society?

COMMUNICATION PRINCIPLES

How do engineers build evidence-based arguments?

Connection from last class

Last week: research, design, and innovation create technological possibilities.

Video calling existed before it became part of everyday life.

TECHNOLOGY

What had to exist for people to use video calling?

CONTEXT

What changed when work and learning moved online?

DIFFUSION

How did other people's use influence adoption?

Chat or discuss: What changed around the technology?

Why might an innovation spread?

Relative advantage

Better for whom, compared with what?

Complexity

How difficult does it seem to understand or use?

Observability

Can people see its use and results?

Compatibility

Does it fit existing routines, values, and systems?

Trialability

Can users try it before committing?

Your example

Which factor helps explain a technology you use?

Choose one factor. Explain how it could encourage or discourage adoption.

Same vehicle, different adoption decisions

Fictional scenario: two people consider the same EV at the same price. Both have similar daily commutes.

Alex has home charging. Sam rents an apartment with no charging access.

- **Which Rogers characteristic helps explain their different decisions?**
- **What wider systems shape those decisions?**
- **What could make adoption more practical for Sam?**
- **Would waiting necessarily mean Sam dislikes new technology?**

Think about housing, charging infrastructure, standards, cost, and policy.

Generative AI: adoption within a system



Choose one wider force. How could it encourage adoption or create pressure to adopt?

Does rapid diffusion tell us whether a technology is accurate, fair, safe, or beneficial?

COMMUNICATION PRINCIPLES

How do engineers build evidence-based arguments?

A recommendation needs evidence and an explanation of why that evidence supports it.

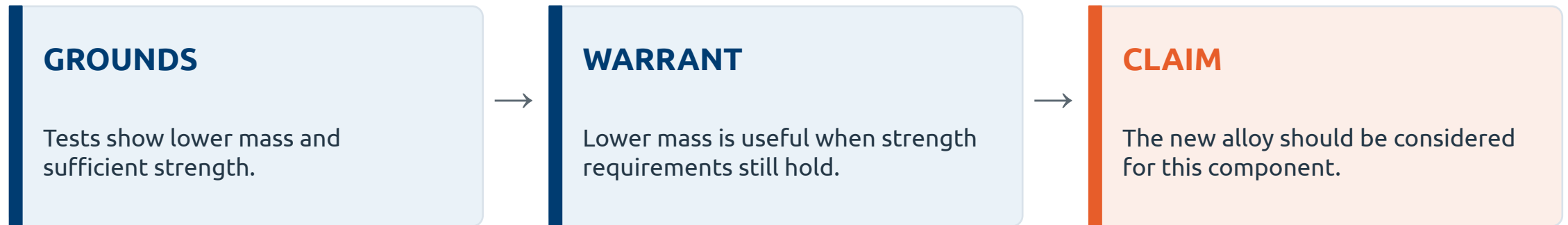
A claim with evidence: enough?

“The new alloy is the better choice because it is lighter.”

What would you need to know before accepting this recommendation?

- What does “better” mean for this component?
- How much lighter, and according to which test?
- Does it meet the required performance criteria?
- Why should reduced mass determine the choice?
- What costs or limits could change the recommendation?

The core of an argument



Which assumption connects the evidence to the recommendation?

- Is reducing mass a priority for this project?
- Does the test reflect the intended use?
- Could another requirement matter more?

QUICK CHECK

If low cost is the main priority, would the same evidence justify the same recommendation?

Context and limits strengthen the argument

BACKING: WHY THIS REASONING FITS

Project specifications make reducing mass a priority, provided minimum strength and safety requirements still hold.

How does this support the warrant?

What else could provide backing: a standard, engineering principle, or design requirement?

QUALIFIER AND REBUTTAL

Qualifier: "Based on the current tests, the alloy appears suitable for this component."

Rebuttal: "Its higher cost could favour the current material if cost becomes the main priority."

Which limits confidence? Which could change the recommendation?

How can these details make a recommendation more defensible?

Your turn: identify the argument parts

Match A–F to claim, grounds, warrant, backing, qualifier, and rebuttal.

Label	Statement
A	Project specifications prioritize lower mass while preserving strength and safety.
B	Higher cost could favour the current material if cost becomes the main priority.
C	Tests found 18% lower mass while meeting minimum strength requirements.
D	The alloy appears suitable based on the current tests.
E	The team should consider the new alloy.
F	Lower mass is an advantage when required strength still holds.

Your turn: repair an overclaim

“Because many people use generative AI, every engineering team should adopt it.”

CLAIM

What narrower recommendation could you defend?

GROUND

What evidence would you need about the intended task?

WARRANT

Why would that evidence justify the recommendation?

BACKING

Which requirements or principles support the reasoning?

QUALIFIER

What conditions or uncertainty would you state?

REBUTTAL

What exception or trade-off could change the conclusion?

Discuss: What does widespread use show, and what does it leave unanswered?

Exit-ticket case: digital inspection records

PILOT

12 inspectors tested the app for 3 weeks.

TIME

Average record time: 14 minutes with the app, 20 on paper.

QUALITY

The pilot found similar rates of missing required fields.

CONTEXT

Site A has tablets and Wi-Fi. Site B has neither.

PRIORITY

The project requires complete records and aims to reduce recording time.

Fictional practice data. Costs, long-term reliability, and offline use remain untested.

Decision: expand at Site A, extend the pilot, or wait? Explain your choice.

Exit ticket: adoption and a defensible claim

With your group, use the digital inspection case. About 120–160 words total.

1. EXPLAIN ADOPTION

Name one Rogers characteristic that helps explain adoption at Site A or Site B.

Explain how one wider system condition affects adoption at the two sites.

Use details from the case.

2. WRITE A RECOMMENDATION

Write a short paragraph with:

A clear claim and relevant pilot evidence.
Reasoning that links the evidence to the claim.
The project requirement supporting that reasoning.
A qualifier and one meaningful limit or exception.

Write a connected explanation. Work together, then submit your own copy.

Breakout Room Instructions

Work together. Submit individually.

One person opens a blank Google Doc.
Share it with group members as editors.
Copy in the case and prompts from the slides.
Work together on your response.
Each student downloads a copy as Microsoft Word (.docx).
Rename it with your own name and submit it to Canvas.

**Before leaving your breakout room:
make sure every person has
downloaded their own copy.**

Use the case and prompts on the previous two slides

Before you go

TECHNOLOGY & SOCIETY

Adoption depends on how people perceive an innovation and on the systems that make using it possible. Widespread use alone does not establish good outcomes.

COMMUNICATION PRINCIPLES

An argument connects a claim to evidence through reasoning. Backing supports that reasoning. Qualifiers and rebuttals communicate its scope and limits.

Coming next: Technological inequality and the digital divide, with a closer look at reasoning and rhetoric.